Design Track

Interview Preparation Series • Professional Edition

LLD Problems — Coded Solutions

Compiled with LaTeX • June 17, 2026

harsha.somisetty01@gmail.com

In This Edition

> **How to use this file:** Each ## section is a self-contained LLD problem. Read Requirements + Class Diagram first, then study the code. Run any block directly with `python3`. The closing sections distill the cross-cutting patterns for rapid revision.

1 The Machine-Coding Approach

1.1 Time-box strategy (45--90 min round)

Phase	Time	Goal
Clarify requirements	5 min	Pin scope; avoid building the wrong thing
Extract entities & relationships	5 min	Noun-verb extraction on whiteboard
Sketch class diagram	5–10 min	Get agreement before coding
Code core classes + happy path	20–30 min	Must compile/run
Add edge cases + extensibility	10 min	Show you think beyond happy path
Demo driver	5 min	Prove it works end-to-end

1.2 What interviewers grade (roughly in priority order)

1. **Working, runnable code** — broken code is an automatic downgrade.
2. **Clean abstractions** — classes with single responsibilities; meaningful names.
3. **Extensibility** — adding a new feature should require adding code, not changing existing code (OCP).
4. **Edge cases handled** — not just happy path (full parking lot, divide-by-zero in splits, empty elevator queue).
5. **Design patterns named** — shows vocabulary; don't force patterns but do name what you use.

1.3 What to deprioritize under time pressure

- Persistence / DB layer (mention it, skip it).
- Concurrency / thread-safety (mention locks, skip implementation unless asked).
- Perfect error messages (simple exceptions are fine).
- Over-engineering: 3-layer factory + abstract-factory combos for a 45-min round are a red flag.

1.4 The noun-verb extraction recipe

From the problem statement: circle **nouns** → candidate classes; underline **verbs** → candidate methods. Then prune: nouns that are just attributes (e.g. ”name”, ”price”) become fields, not classes.

2 1. LRU Cache (+ LFU Variant Note)

2.1 Requirements

Functional

- `get(key)` — return value or -1 if not present.
- `put(key, value)` — insert; if capacity exceeded, evict **least recently used** entry.

- Both operations must be **O(1)**.

Clarifying questions to ask

- Is capacity fixed at construction? (Yes.)
- Thread-safe? (Assume single-threaded unless asked; mention `threading.Lock` for MT.)
- Key/value types? (Generics / `Any` for flexibility.)
- On `put` of existing key, update value and recency? (Yes.)

2.2 Core entities / classes

Class	Responsibility
`DLinkedNode`	Doubly-linked list node (key, value, prev, next pointers)
`LRUCache`	HashMap + doubly-linked list; exposes `get/put`

2.3 Class diagram (ASCII UML)

```
+------------------+          +-------------------+
|   DLinkedNode    |          |    LRUCache       |
+------------------+          +-------------------+
| key: int         |  1    *  | capacity: int     |
| value: int       |<-------->| cache: dict       |
| prev: DLinkedNode|          | head: DLinkedNode |
| next: DLinkedNode|          | tail: DLinkedNode |
+------------------+          +-------------------+
                              | get(key) -> int   |
                              | put(key, val)     |
                              | _remove(node)     |
                              | _add_to_front(n)  |
                              +-------------------+
```

2.4 Design patterns used

- **No classic GoF pattern** — this is a pure data-structure problem. The key insight is the **sentinel head/tail** trick to avoid null checks. Interviewers want to see you know this.
- Mention: for thread-safety you'd wrap with a `threading.Lock` or use `collections.OrderedDict` (which Python's stdlib provides for a simpler implementation, but interviewers want the manual version to test understanding).

2.5 Full code

```python
from __future__ import annotations
from typing import Optional


class DLinkedNode:
    """Doubly-linked list node used as the internal building block."""

    def __init__(self, key: int = 0, value: int = 0) -> None:
        self.key = key
        self.value = value
        self.prev: Optional[DLinkedNode] = None
        self.next: Optional[DLinkedNode] = None
```

```python


class LRUCache:
    """
    O(1) get/put LRU Cache implemented with a HashMap + doubly-linked list.

    Invariant: the list is ordered most-recently-used (front) to
    least-recently-used (back). Sentinel head/tail nodes eliminate
    edge-case null checks.
    """

    def __init__(self, capacity: int) -> None:
        if capacity <= 0:
            raise ValueError("Capacity must be positive")
        self.capacity = capacity
        self.cache: dict[int, DLinkedNode] = {}

        # Sentinels — never evicted, never returned
        self.head = DLinkedNode()  # MRU sentinel
        self.tail = DLinkedNode()  # LRU sentinel
        self.head.next = self.tail
        self.tail.prev = self.head

    # ------------------------------------------------------------------ #
    # Public API                                                          #
    # ------------------------------------------------------------------ #

    def get(self, key: int) -> int:
        if key not in self.cache:
            return -1
        node = self.cache[key]
        self._move_to_front(node)
        return node.value

    def put(self, key: int, value: int) -> None:
        if key in self.cache:
            node = self.cache[key]
            node.value = value
            self._move_to_front(node)
        else:
            node = DLinkedNode(key, value)
            self.cache[key] = node
            self._add_to_front(node)
            if len(self.cache) > self.capacity:
                lru = self._pop_from_back()
                del self.cache[lru.key]

    # ------------------------------------------------------------------ #
    # Private helpers                                                     #
    # ------------------------------------------------------------------ #

    def _add_to_front(self, node: DLinkedNode) -> None:
        """Insert node right after head sentinel (= most-recently-used)."""
        node.prev = self.head
        node.next = self.head.next
        self.head.next.prev = node
        self.head.next = node

    def _remove(self, node: DLinkedNode) -> None:
        """Unlink node from its current position."""
        node.prev.next = node.next
```

```
        node.next.prev = node.prev

    def _move_to_front(self, node: DLinkedNode) -> None:
        self._remove(node)
        self._add_to_front(node)

    def _pop_from_back(self) -> DLinkedNode:
        """Remove and return the LRU node (just before tail sentinel)."""
        lru = self.tail.prev
        self._remove(lru)
        return lru


# ---------------------------------------------------------------------- #
# LFU variant — brief notes (full implementation would add freq map)     #
# ---------------------------------------------------------------------- #
# LFU Cache (Least Frequently Used):
#   - Evict the entry with the LOWEST access frequency;
#     tie-break on LRU order within same frequency.
# Data structures needed:
#   - key_to_val: dict[key → value]
#   - key_to_freq: dict[key → frequency]
#   - freq_to_keys: dict[freq → OrderedDict[key → None]]  (ordered = LRU within freq)
#   - min_freq: int  (track current minimum)
# On get/put: increment freq, move key to freq+1 bucket, update min_freq.
# All operations O(1). See Python's collections.OrderedDict for ordered bucket.


if __name__ == "__main__":
    cache = LRUCache(capacity=3)
    cache.put(1, 10)
    cache.put(2, 20)
    cache.put(3, 30)

    print(cache.get(1))   # 10  — 1 is now MRU
    cache.put(4, 40)      # evicts 2 (LRU)
    print(cache.get(2))   # -1  — evicted
    print(cache.get(3))   # 30
    cache.put(5, 50)      # evicts 4 (LRU after 1 and 3 were accessed)
    print(cache.get(4))   # -1
    print(cache.get(1))   # 10
    print(cache.get(3))   # 30
    print(cache.get(5))   # 50
```

2.6 Extensibility discussion

- **TTL / expiry:** Add `expiry: Optional[float]` to `DLinkedNode`. On `get`, check `time.time() > expiry` and treat as miss + remove. A background thread or lazy expiry on access both work.
- **Thread-safety:** Wrap `get`/`put` with `threading.Lock`. For higher concurrency, shard the cache into N sub-caches keyed by `hash(key) % N`.
- **Generics:** Replace `int` with `TypeVar('K')` / `TypeVar('V')` and reuse across types.

Interview takeaway: The sentinel head/tail trick is the key insight. Without sentinels you need `if node.prev` null-checks everywhere — messy and error-prone. Name it explicitly.

3 2. Parking Lot

3.1 Requirements

Functional

- Support multiple levels, multiple spot types (COMPACT, LARGE, MOTORCYCLE, ELECTRIC).
- Vehicles: Motorcycle, Car, Truck — each fits specific spot types.
- `park(vehicle)` — find nearest available spot; return ticket.
- `unpark(ticket)` — free the spot; compute fee.
- Pluggable pricing strategy (hourly, flat-rate, etc.).
- Display available spots per level.

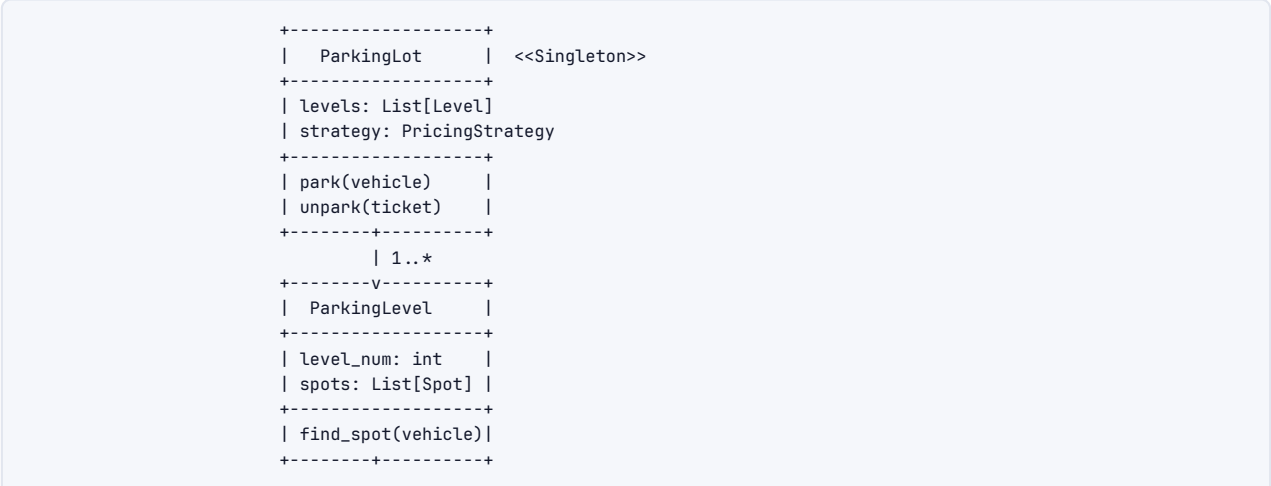
Clarifying questions to ask

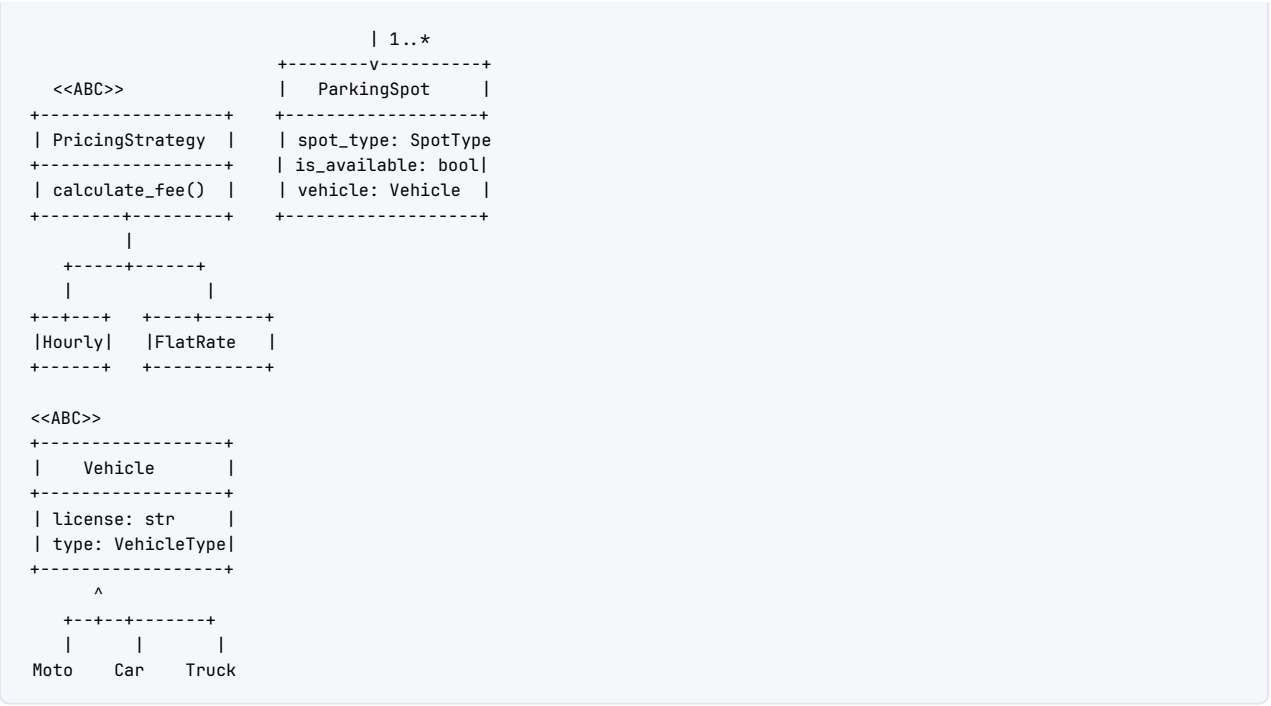
- Can a car park in a large spot if compact is full? (Yes, with preference ordering.)
- Is pricing per hour or per entry? (Make it a strategy — both.)
- Multiple entrances? (Assume single for now; mention extension.)
- Reservations / pre-booking? (Out of scope.)
- Thread-safety for concurrent `park` calls? (Mention lock per level; skip for now.)

3.2 Core entities / classes

Class	Responsibility
`VehicleType` (enum)	MOTORCYCLE, CAR, TRUCK
`SpotType` (enum)	MOTORCYCLE, COMPACT, LARGE, ELECTRIC
`Vehicle` (ABC)	Base vehicle with type + license plate
`ParkingSpot`	Individual spot — type, level, number, occupied flag
`ParkingLevel`	Collection of spots on one floor; find-spot logic
`ParkingLot`	Singleton; owns levels; exposes park/unpark
`Ticket`	Issued on park; holds spot ref + entry time
`PricingStrategy` (ABC)	Strategy interface: `calculate_fee(ticket) → float`
`HourlyPricing`	Concrete strategy
`FlatRatePricing`	Concrete strategy

3.3 Class diagram (ASCII UML)





3.4 Design patterns used

- **Strategy** — `PricingStrategy` lets you swap hourly/flat-rate/weekend pricing without touching `ParkingLot`.
- **Singleton** — `ParkingLot` represents a single physical lot; use `__new__` or a module-level instance.
- **Factory Method** — `VehicleFactory.create(type, plate)` to instantiate vehicles without `if/else` in client code.
- **Template Method** — `Vehicle` defines `allowed_spot_types()` as an abstract method; each subclass overrides it.

3.5 Full code

Python

```python
from __future__ import annotations

import time
from abc import ABC, abstractmethod
from enum import Enum, auto
from typing import Optional
from dataclasses import dataclass, field


# ------------------------------------------------------------------ #
# Enums                                                               #
# ------------------------------------------------------------------ #

class VehicleType(Enum):
    MOTORCYCLE = auto()
    CAR = auto()
    TRUCK = auto()


class SpotType(Enum):
    MOTORCYCLE = auto()
```

```
    COMPACT = auto()
    LARGE = auto()
    ELECTRIC = auto()


# ---------------------------------------------------------------- #
# Vehicles                                                         #
# ---------------------------------------------------------------- #

class Vehicle(ABC):
    def __init__(self, license_plate: str, vehicle_type: VehicleType) -> None:
        self.license_plate = license_plate
        self.vehicle_type = vehicle_type

    @abstractmethod
    def allowed_spot_types(self) -> list[SpotType]:
        """Ordered preference: first = most preferred spot type."""
        ...

    def __repr__(self) -> str:
        return f"{self.__class__.__name__}({self.license_plate})"


class Motorcycle(Vehicle):
    def __init__(self, license_plate: str) -> None:
        super().__init__(license_plate, VehicleType.MOTORCYCLE)

    def allowed_spot_types(self) -> list[SpotType]:
        return [SpotType.MOTORCYCLE, SpotType.COMPACT, SpotType.LARGE]


class Car(Vehicle):
    def __init__(self, license_plate: str) -> None:
        super().__init__(license_plate, VehicleType.CAR)

    def allowed_spot_types(self) -> list[SpotType]:
        return [SpotType.COMPACT, SpotType.LARGE, SpotType.ELECTRIC]


class Truck(Vehicle):
    def __init__(self, license_plate: str) -> None:
        super().__init__(license_plate, VehicleType.TRUCK)

    def allowed_spot_types(self) -> list[SpotType]:
        return [SpotType.LARGE]


class VehicleFactory:
    """Factory to decouple client from concrete vehicle classes."""

    _registry: dict[VehicleType, type[Vehicle]] = {
        VehicleType.MOTORCYCLE: Motorcycle,
        VehicleType.CAR: Car,
        VehicleType.TRUCK: Truck,
    }

    @classmethod
    def create(cls, vehicle_type: VehicleType, license_plate: str) -> Vehicle:
        klass = cls._registry.get(vehicle_type)
        if klass is None:
            raise ValueError(f"Unknown vehicle type: {vehicle_type}")
```

```
        return klass(license_plate)


# ---------------------------------------------------------------------- #
# Parking Spot                                                           #
# ---------------------------------------------------------------------- #

class ParkingSpot:
    def __init__(self, spot_id: str, spot_type: SpotType, level: int) -> None:
        self.spot_id = spot_id
        self.spot_type = spot_type
        self.level = level
        self._vehicle: Optional[Vehicle] = None

    @property
    def is_available(self) -> bool:
        return self._vehicle is None

    def park(self, vehicle: Vehicle) -> None:
        if not self.is_available:
            raise RuntimeError(f"Spot {self.spot_id} already occupied")
        self._vehicle = vehicle

    def unpark(self) -> Vehicle:
        if self.is_available:
            raise RuntimeError(f"Spot {self.spot_id} is empty")
        vehicle = self._vehicle
        self._vehicle = None
        return vehicle  # type: ignore[return-value]

    def __repr__(self) -> str:
        status = "FREE" if self.is_available else f"OCC({self._vehicle})"
        return f"Spot[{self.spot_id},{self.spot_type.name},{status}]"


# ---------------------------------------------------------------------- #
# Ticket                                                                 #
# ---------------------------------------------------------------------- #

@dataclass
class Ticket:
    ticket_id: str
    vehicle: Vehicle
    spot: ParkingSpot
    entry_time: float = field(default_factory=time.time)
    exit_time: Optional[float] = None

    def duration_hours(self) -> float:
        end = self.exit_time or time.time()
        return (end - self.entry_time) / 3600.0


# ---------------------------------------------------------------------- #
# Pricing Strategy                                                       #
# ---------------------------------------------------------------------- #

class PricingStrategy(ABC):
    @abstractmethod
    def calculate_fee(self, ticket: Ticket) -> float:
        ...

```

```python


class HourlyPricing(PricingStrategy):
    def __init__(self, rate_per_hour: float = 2.0) -> None:
        self.rate_per_hour = rate_per_hour

    def calculate_fee(self, ticket: Ticket) -> float:
        import math
        hours = math.ceil(ticket.duration_hours())  # round up to next hour
        return max(hours, 1) * self.rate_per_hour


class FlatRatePricing(PricingStrategy):
    def __init__(self, flat_fee: float = 5.0) -> None:
        self.flat_fee = flat_fee

    def calculate_fee(self, ticket: Ticket) -> float:
        return self.flat_fee


# ---------------------------------------------------------------------- #
# Parking Level                                                          #
# ---------------------------------------------------------------------- #

class ParkingLevel:
    def __init__(self, level_num: int, spots: list[ParkingSpot]) -> None:
        self.level_num = level_num
        self.spots = spots

    def find_spot(self, vehicle: Vehicle) -> Optional[ParkingSpot]:
        """Return first available spot matching vehicle preference order."""
        for preferred_type in vehicle.allowed_spot_types():
            for spot in self.spots:
                if spot.spot_type == preferred_type and spot.is_available:
                    return spot
        return None

    def available_count(self) -> dict[SpotType, int]:
        counts: dict[SpotType, int] = {}
        for spot in self.spots:
            if spot.is_available:
                counts[spot.spot_type] = counts.get(spot.spot_type, 0) + 1
        return counts


# ---------------------------------------------------------------------- #
# Parking Lot (Singleton)                                                #
# ---------------------------------------------------------------------- #

class ParkingLot:
    _instance: Optional[ParkingLot] = None

    def __new__(cls, *args, **kwargs) -> ParkingLot:
        if cls._instance is None:
            cls._instance = super().__new__(cls)
        return cls._instance

    def __init__(
        self,
        name: str,
        levels: list[ParkingLevel],
        pricing_strategy: Optional[PricingStrategy] = None,
```

```
    ) -> None:
        # Guard against re-init on subsequent __new__ returns
        if hasattr(self, "_initialized"):
            return
        self.name = name
        self.levels = levels
        self.pricing_strategy: PricingStrategy = pricing_strategy or HourlyPricing()
        self._tickets: dict[str, Ticket] = {}
        self._ticket_counter = 0
        self._initialized = True

    def park(self, vehicle: Vehicle) -> Ticket:
        for level in self.levels:
            spot = level.find_spot(vehicle)
            if spot:
                spot.park(vehicle)
                self._ticket_counter += 1
                ticket = Ticket(
                    ticket_id=f"T{self._ticket_counter:04d}",
                    vehicle=vehicle,
                    spot=spot,
                )
                self._tickets[ticket.ticket_id] = ticket
                print(f"  Parked {vehicle} at {spot.spot_id} (Level {level.level_num})")
                return ticket
        raise RuntimeError("No available spot for this vehicle type")

    def unpark(self, ticket_id: str) -> float:
        ticket = self._tickets.get(ticket_id)
        if ticket is None:
            raise ValueError(f"Unknown ticket: {ticket_id}")
        ticket.exit_time = time.time()
        ticket.spot.unpark()
        fee = self.pricing_strategy.calculate_fee(ticket)
        del self._tickets[ticket_id]
        print(f"  Unparked {ticket.vehicle} | Fee: ${fee:.2f}")
        return fee

    def display_availability(self) -> None:
        print(f"\n--- {self.name} Availability ---")
        for level in self.levels:
            counts = level.available_count()
            print(f"  Level {level.level_num}: {counts}")


# ------------------------------------------------------------------ #
# Helper: build a lot                                                 #
# ------------------------------------------------------------------ #

def build_parking_lot() -> ParkingLot:
    # Reset singleton for demo purposes
    ParkingLot._instance = None

    def make_spots(level: int, config: dict[SpotType, int]) -> list[ParkingSpot]:
        spots = []
        for spot_type, count in config.items():
            for i in range(1, count + 1):
                spot_id = f"L{level}-{spot_type.name[0]}{i}"
                spots.append(ParkingSpot(spot_id, spot_type, level))
        return spots

```

```
    levels = [
        ParkingLevel(1, make_spots(1, {
            SpotType.MOTORCYCLE: 5,
            SpotType.COMPACT: 10,
            SpotType.LARGE: 5,
        })),
        ParkingLevel(2, make_spots(2, {
            SpotType.COMPACT: 8,
            SpotType.LARGE: 4,
            SpotType.ELECTRIC: 3,
        })),
    ]
    return ParkingLot("City Center Garage", levels, HourlyPricing(rate_per_hour=3.0))


if __name__ == "__main__":
    lot = build_parking_lot()
    lot.display_availability()

    moto = Motorcycle("MOTO-001")
    car1 = Car("CAR-001")
    truck = Truck("TRK-001")

    t1 = lot.park(moto)
    t2 = lot.park(car1)
    t3 = lot.park(truck)
    lot.display_availability()

    # Simulate time passing (normally you'd actually wait)
    t2.entry_time -= 7200  # pretend car parked 2 hours ago
    lot.unpark(t2.ticket_id)
    lot.unpark(t1.ticket_id)
    lot.display_availability()
```

3.6 Extensibility discussion

- **New vehicle type (e.g., Bus):** Add `VehicleType.BUS`, create `class Bus(Vehicle)` with `allowed_spot_types`, register in `VehicleFactory._registry`. Zero changes to `ParkingLot` or `ParkingLevel`.
- **New pricing (weekend/dynamic):** Implement `WeekendPricing(PricingStrategy)`. Pass it at construction. OCP satisfied.
- **EV charging:** Add `SpotType.ELECTRIC`; `Car.allowed_spot_types` already supports it. Add `ChargeSession` attached to `Ticket`.
- **Reservations:** Add `reserve(vehicle, start_time)` that marks a spot with a future timestamp; `find_spot` skips reserved spots for the overlapping window.

Interview takeaway: Always name the Strategy pattern for pricing — it's the most common follow-up ("what if pricing changes on weekends?"). The answer is: new class, zero changes to existing code.

4 3. Rate Limiter

4.1 Requirements

Functional

- `allow(client_id)` → `bool` — return True if request should be permitted.

- Support Token Bucket, Sliding Window Log, Fixed Window Counter strategies.
- Pluggable — caller picks strategy at construction.
- Per-client limiting (different clients independent).

Clarifying questions to ask

- Global limit or per-client? (Per-client.)
- Burst tolerance? (Token Bucket allows bursts; fixed-window does not.)
- Distributed (Redis-backed)? (Assume in-process for now; mention Redis extension.)
- What happens on limit hit — queue or reject? (Reject / return False.)
- Thread-safe? (Yes, per-client locks.)

4.2 Core entities / classes

Class	Responsibility
`RateLimitStrategy` (ABC)	Interface: `is_allowed(client_id) → bool`
`TokenBucketStrategy`	Refill tokens at fixed rate; allow if tokens > 0
`FixedWindowStrategy`	Count requests per fixed time window; reset on window roll
`SlidingWindowLogStrategy`	Keep timestamped log; count entries in last N seconds
`RateLimiter`	Context object; delegates to strategy

4.3 Class diagram (ASCII UML)

```
+----------------------+
|     RateLimiter      |
+----------------------+
| strategy: RateLimitStrategy
+----------------------+
| allow(client_id)→bool|
+----------+-----------+
           |
           | uses
+----------v-----------+       <<ABC>>
| RateLimitStrategy    |<----------------------+
+----------------------+                       |
| is_allowed(cid)→bool|          +-------------+-------------+
+----------------------+          |             |             |
                        TokenBucket   FixedWindow   SlidingWindowLog
```

4.4 Design patterns used

- **Strategy** — core pattern; swap algorithms at runtime.
- **Factory / constructor injection** — `RateLimiter(strategy=TokenBucketStrategy(...))`.

4.5 Full code

```python
from __future__ import annotations

import time
import threading
from abc import ABC, abstractmethod
```

```python
from collections import deque
from dataclasses import dataclass, field
from typing import Optional


# ---------------------------------------------------------------------- #
# Strategy Interface                                                     #
# ---------------------------------------------------------------------- #

class RateLimitStrategy(ABC):
    @abstractmethod
    def is_allowed(self, client_id: str) -> bool:
        ...


# ---------------------------------------------------------------------- #
# Strategy 1: Token Bucket                                               #
# ---------------------------------------------------------------------- #

@dataclass
class _TokenBucketState:
    tokens: float
    last_refill: float = field(default_factory=time.time)


class TokenBucketStrategy(RateLimitStrategy):
    """
    Each client starts with `capacity` tokens.
    Tokens refill at `refill_rate` tokens/second.
    Each request consumes 1 token. Allows bursts up to `capacity`.
    """

    def __init__(self, capacity: float, refill_rate: float) -> None:
        self.capacity = capacity
        self.refill_rate = refill_rate
        self._buckets: dict[str, _TokenBucketState] = {}
        self._lock = threading.Lock()

    def _get_bucket(self, client_id: str) -> _TokenBucketState:
        if client_id not in self._buckets:
            self._buckets[client_id] = _TokenBucketState(tokens=self.capacity)
        return self._buckets[client_id]

    def is_allowed(self, client_id: str) -> bool:
        with self._lock:
            now = time.time()
            bucket = self._get_bucket(client_id)
            elapsed = now - bucket.last_refill
            bucket.tokens = min(
                self.capacity,
                bucket.tokens + elapsed * self.refill_rate
            )
            bucket.last_refill = now
            if bucket.tokens >= 1:
                bucket.tokens -= 1
                return True
            return False


# ---------------------------------------------------------------------- #
# Strategy 2: Fixed Window Counter                                       #
```

```python
# ---------------------------------------------------------------- #

@dataclass
class _FixedWindowState:
    count: int = 0
    window_start: float = field(default_factory=time.time)


class FixedWindowStrategy(RateLimitStrategy):
    """
    Allow up to `limit` requests per `window_seconds` window.
    Window resets at the start of each period.
    Weakness: burst at window boundary (2x limit in 2*epsilon seconds).
    """

    def __init__(self, limit: int, window_seconds: float) -> None:
        self.limit = limit
        self.window_seconds = window_seconds
        self._states: dict[str, _FixedWindowState] = {}
        self._lock = threading.Lock()

    def is_allowed(self, client_id: str) -> bool:
        with self._lock:
            now = time.time()
            if client_id not in self._states:
                self._states[client_id] = _FixedWindowState(window_start=now)
            state = self._states[client_id]

            if now - state.window_start >= self.window_seconds:
                state.count = 0
                state.window_start = now

            if state.count < self.limit:
                state.count += 1
                return True
            return False


# ---------------------------------------------------------------- #
# Strategy 3: Sliding Window Log                                   #
# ---------------------------------------------------------------- #

class SlidingWindowLogStrategy(RateLimitStrategy):
    """
    Keep a deque of request timestamps per client.
    On each request, prune entries older than `window_seconds`,
    then check if remaining count < limit.
    Most accurate; memory = O(limit) per client.
    """

    def __init__(self, limit: int, window_seconds: float) -> None:
        self.limit = limit
        self.window_seconds = window_seconds
        self._logs: dict[str, deque[float]] = {}
        self._lock = threading.Lock()

    def is_allowed(self, client_id: str) -> bool:
        with self._lock:
            now = time.time()
            if client_id not in self._logs:
                self._logs[client_id] = deque()
```

```
            log = self._logs[client_id]

            cutoff = now - self.window_seconds
            while log and log[0] < cutoff:
                log.popleft()

            if len(log) < self.limit:
                log.append(now)
                return True
            return False


# ---------------------------------------------------------------------- #
# Rate Limiter Context                                                   #
# ---------------------------------------------------------------------- #

class RateLimiter:
    """
    Context class. Delegates to the injected strategy.
    Swap strategy at runtime with `set_strategy`.
    """

    def __init__(self, strategy: RateLimitStrategy) -> None:
        self._strategy = strategy

    def set_strategy(self, strategy: RateLimitStrategy) -> None:
        self._strategy = strategy

    def allow(self, client_id: str) -> bool:
        return self._strategy.is_allowed(client_id)


# ---------------------------------------------------------------------- #
# Demo                                                                   #
# ---------------------------------------------------------------------- #

if __name__ == "__main__":
    print("=== Token Bucket (capacity=5, refill=1/s) ===")
    rl = RateLimiter(TokenBucketStrategy(capacity=5, refill_rate=1.0))
    for i in range(7):
        result = rl.allow("user_A")
        print(f"  Request {i+1}: {'ALLOWED' if result else 'DENIED'}")

    print("\n=== Fixed Window (limit=3, window=10s) ===")
    rl2 = RateLimiter(FixedWindowStrategy(limit=3, window_seconds=10))
    for i in range(5):
        result = rl2.allow("user_B")
        print(f"  Request {i+1}: {'ALLOWED' if result else 'DENIED'}")

    print("\n=== Sliding Window Log (limit=3, window=1s) ===")
    rl3 = RateLimiter(SlidingWindowLogStrategy(limit=3, window_seconds=1.0))
    for i in range(4):
        result = rl3.allow("user_C")
        print(f"  Request {i+1}: {'ALLOWED' if result else 'DENIED'}")
    time.sleep(1.1)
    result = rl3.allow("user_C")
    print(f"  After sleep: {'ALLOWED' if result else 'DENIED'}")
```

4.6 Extensibility discussion

- **Redis-backed distributed limiter:** Replace the in-memory `_buckets`/`_logs` dicts with Redis calls using atomic `INCR`/`EXPIRE` or Lua scripts. Strategy interface unchanged.
- **New algorithm (leaky bucket):** Implement `LeakyBucketStrategy(RateLimitStrategy)`. Inject it. No other changes.
- **Middleware integration:** Wrap `RateLimiter.allow` in a decorator or WSGI/ASGI middleware. The core class is unaware of HTTP.
- **Per-endpoint limits:** Key on `f"{client_id}:{endpoint}"` instead of just `client_id`.

Interview takeaway: Know the trade-offs: Token Bucket allows bursts; Fixed Window has boundary-burst weakness; Sliding Window Log is most accurate but costs O(limit) memory per client. Sliding Window Counter (hybrid) is the production sweet spot.

5 4. Splitwise / Expense Sharing

5.1 Requirements

Functional

- Add users to groups.
- Add an expense: payer, amount, split type (EQUAL, EXACT, PERCENT), participants.
- `show_balances(user)` — who owes whom and how much.
- `simplify_debts(group)` — minimize number of transactions to settle.

Clarifying questions to ask

- Currency? (Single currency; mention multi-currency extension.)
- Can the payer be a participant? (Yes — they effectively owe zero or receive money.)
- Simplify debts: across whole app or per-group? (Per-group for now.)
- Partial payments / settle-up? (Yes — add expense with type = SETTLE.)
- Persistence? (In-memory; mention DB extension.)

5.2 Core entities / classes

Class	Responsibility
`User`	Name, id, personal balance map
`Group`	Collection of users + expenses
`Expense`	Amount, payer, participants, split strategy
`SplitStrategy` (ABC)	`compute_shares(amount, participants) → dict[User, float]`
`EqualSplit`	Divide evenly
`ExactSplit`	Caller specifies each person's share
`PercentSplit`	Caller specifies each person's percentage
`Balance`	Utility: compute net balances + simplify

5.3 Class diagram (ASCII UML)

```
+----------+       +----------+       +------------+
|  User    |       |  Group   |       |  Expense   |
+----------+       +----------+       +------------+
| id: str  |<----->| users    |<------| amount     |
| name: str|       | expenses |       | payer: User|
+----------+       +----------+       | split: Split
                                      +-----+------+
                                            | uses
                              <<ABC>>  +----v------+
                          +-----------+|SplitStrat |
                          |  Equal    |+-----------+
                          |  Exact    || compute() |
                          |  Percent  |+-----------+
                          +-----------+
```

5.4 Design patterns used

- **Strategy** — `SplitStrategy` for different split computations.
- **Domain Model** — `User`, `Group`, `Expense` map directly to business entities.

5.5 Full code

```python
from __future__ import annotations

from abc import ABC, abstractmethod
from dataclasses import dataclass, field
from enum import Enum, auto
from typing import Optional
import uuid


# ---------------------------------------------------------------------- #
# Split Strategies                                                       #
# ---------------------------------------------------------------------- #

class SplitStrategy(ABC):
    @abstractmethod
    def compute_shares(
        self, amount: float, participants: list[User], **kwargs
    ) -> dict[str, float]:
        """Return {user_id: share_amount} for each participant."""
        ...


class EqualSplit(SplitStrategy):
    def compute_shares(
        self, amount: float, participants: list[User], **kwargs
    ) -> dict[str, float]:
        share = round(amount / len(participants), 2)
        result = {u.user_id: share for u in participants}
        # Correct rounding error on first participant
        total = sum(result.values())
        diff = round(amount - total, 2)
        first = participants[0].user_id
        result[first] = round(result[first] + diff, 2)
```

```python
        return result


class ExactSplit(SplitStrategy):
    def compute_shares(
        self, amount: float, participants: list[User], **kwargs
    ) -> dict[str, float]:
        shares: dict[str, float] = kwargs.get("exact_shares", {})
        if abs(sum(shares.values()) - amount) > 0.01:
            raise ValueError("Exact shares must sum to total amount")
        return {u.user_id: shares[u.user_id] for u in participants}


class PercentSplit(SplitStrategy):
    def compute_shares(
        self, amount: float, participants: list[User], **kwargs
    ) -> dict[str, float]:
        percents: dict[str, float] = kwargs.get("percents", {})
        if abs(sum(percents.values()) - 100) > 0.01:
            raise ValueError("Percentages must sum to 100")
        return {
            u.user_id: round(amount * percents[u.user_id] / 100, 2)
            for u in participants
        }


# ---------------------------------------------------------------------- #
# User                                                                   #
# ---------------------------------------------------------------------- #

@dataclass
class User:
    name: str
    user_id: str = field(default_factory=lambda: str(uuid.uuid4())[:8])

    def __hash__(self) -> int:
        return hash(self.user_id)

    def __eq__(self, other: object) -> bool:
        return isinstance(other, User) and self.user_id == other.user_id

    def __repr__(self) -> str:
        return f"User({self.name})"


# ---------------------------------------------------------------------- #
# Expense                                                                #
# ---------------------------------------------------------------------- #

class SplitType(Enum):
    EQUAL = auto()
    EXACT = auto()
    PERCENT = auto()


@dataclass
class Expense:
    expense_id: str
    description: str
    amount: float
    payer: User
```

```
    participants: list[User]
    split_strategy: SplitStrategy
    shares: dict[str, float]  # user_id → amount owed by that user


# ---------------------------------------------------------------- #
# Group                                                            #
# ---------------------------------------------------------------- #

class Group:
    def __init__(self, name: str) → None:
        self.group_id = str(uuid.uuid4())[:8]
        self.name = name
        self.members: list[User] = []
        self.expenses: list[Expense] = []
        # net_balances[a][b] = amount b owes a (positive = b owes a)
        self._net: dict[str, dict[str, float]] = {}

    def add_member(self, user: User) → None:
        if user not in self.members:
            self.members.append(user)
            self._net.setdefault(user.user_id, {})

    def add_expense(
        self,
        description: str,
        amount: float,
        payer: User,
        participants: list[User],
        strategy: SplitStrategy,
        **kwargs,
    ) → Expense:
        if payer not in self.members:
            raise ValueError(f"{payer} is not a group member")
        shares = strategy.compute_shares(amount, participants, **kwargs)
        expense = Expense(
            expense_id=str(uuid.uuid4())[:8],
            description=description,
            amount=amount,
            payer=payer,
            participants=participants,
            split_strategy=strategy,
            shares=shares,
        )
        self.expenses.append(expense)
        self._update_balances(expense)
        return expense

    def _update_balances(self, expense: Expense) → None:
        payer_id = expense.payer.user_id
        for uid, share in expense.shares.items():
            if uid == payer_id:
                continue  # payer doesn't owe themselves
            # uid owes payer_id `share` amount
            self._net.setdefault(payer_id, {})
            self._net.setdefault(uid, {})
            current = self._net[payer_id].get(uid, 0)
            self._net[payer_id][uid] = round(current + share, 2)

    def show_balances(self) → None:
        print(f"\n--- Balances in group '{self.name}' ---")
```

```
        any_balance = False
        for creditor_id, debtors in self._net.items():
            creditor_name = self._user_name(creditor_id)
            for debtor_id, amount in debtors.items():
                net = amount - self._net.get(debtor_id, {}).get(creditor_id, 0)
                if net > 0.01:
                    any_balance = True
                    debtor_name = self._user_name(debtor_id)
                    print(f"  {debtor_name} owes {creditor_name}: ${net:.2f}")
        if not any_balance:
            print("  All settled up!")

    def simplify_debts(self) -> list[tuple[str, str, float]]:
        """
        Minimize number of transactions using a greedy creditor-debtor matching.
        Returns list of (debtor_name, creditor_name, amount).
        """
        # Compute net position for each user
        net_position: dict[str, float] = {m.user_id: 0.0 for m in self.members}
        for creditor_id, debtors in self._net.items():
            for debtor_id, amount in debtors.items():
                net_position[creditor_id] = round(net_position[creditor_id] + amount, 2)
                net_position[debtor_id] = round(net_position[debtor_id] - amount, 2)

        # Separate into creditors (positive) and debtors (negative)
        creditors = {uid: amt for uid, amt in net_position.items() if amt > 0.01}
        debtors = {uid: -amt for uid, amt in net_position.items() if amt < -0.01}

        transactions: list[tuple[str, str, float]] = []
        cred_list = sorted(creditors.items(), key=lambda x: -x[1])
        debt_list = sorted(debtors.items(), key=lambda x: -x[1])

        i, j = 0, 0
        while i < len(cred_list) and j < len(debt_list):
            cid, c_amt = cred_list[i]
            did, d_amt = debt_list[j]
            settled = round(min(c_amt, d_amt), 2)
            transactions.append((self._user_name(did), self._user_name(cid), settled))
            c_amt = round(c_amt - settled, 2)
            d_amt = round(d_amt - settled, 2)
            cred_list[i] = (cid, c_amt)
            debt_list[j] = (d_amt, d_amt)  # dummy; just advance index
            if c_amt < 0.01:
                i += 1
            if d_amt < 0.01:
                j += 1

        return transactions

    def _user_name(self, user_id: str) -> str:
        for m in self.members:
            if m.user_id == user_id:
                return m.name
        return user_id


if __name__ == "__main__":
    alice = User("Alice")
    bob = User("Bob")
    charlie = User("Charlie")
```

```
    group = Group("Trip to Goa")
    for u in [alice, bob, charlie]:
        group.add_member(u)

    # Alice pays Rs 300 split equally
    group.add_expense(
        "Dinner", 300.0, alice, [alice, bob, charlie], EqualSplit()
    )

    # Bob pays Rs 200: Charlie pays 50%, Bob 30%, Alice 20%
    group.add_expense(
        "Cab", 200.0, bob,
        [alice, bob, charlie], PercentSplit(),
        percents={alice.user_id: 20, bob.user_id: 30, charlie.user_id: 50}
    )

    group.show_balances()

    print("\n--- Simplified transactions ---")
    for debtor, creditor, amount in group.simplify_debts():
        print(f"  {debtor} pays {creditor}: ${amount:.2f}")
```

5.6 Extensibility discussion

- **New split type (shares-based):** Add `SharesSplit(SplitStrategy)` where each participant has N shares; their fraction = N_i / sum(N). Zero changes to `Group`.
- **Multi-currency:** Add `currency: str` to `Expense`; convert to a base currency using an injected `CurrencyConverter` before updating balances.
- **Settle-up:** `settle(payer, payee, amount)` is just `add_expense("Settlement", amount, payer, [payer, payee], ExactSplit(), exact_shares={payer.user_id:0, payee.user_id:amount})`.
- **Persistence:** Replace `self._net` updates with DB writes. The `Group` class becomes a service layer calling a repository.

Interview takeaway: The debt-simplification algorithm is a classic greedy: compute net positions, then greedily match biggest creditor with biggest debtor. It's O(N log N) and minimizes the number of transactions (though not necessarily the amounts).

6 5. Elevator System

6.1 Requirements

Functional

- Multiple elevators in a building.
- External requests: `request_floor(floor, direction)` — someone on floor 3 wants to go UP.
- Internal requests: `select_floor(elevator_id, floor)` — someone inside presses a button.
- Scheduler assigns best elevator to each external request.
- Each elevator has states: IDLE, MOVING_UP, MOVING_DOWN, MAINTENANCE.

Clarifying questions to ask

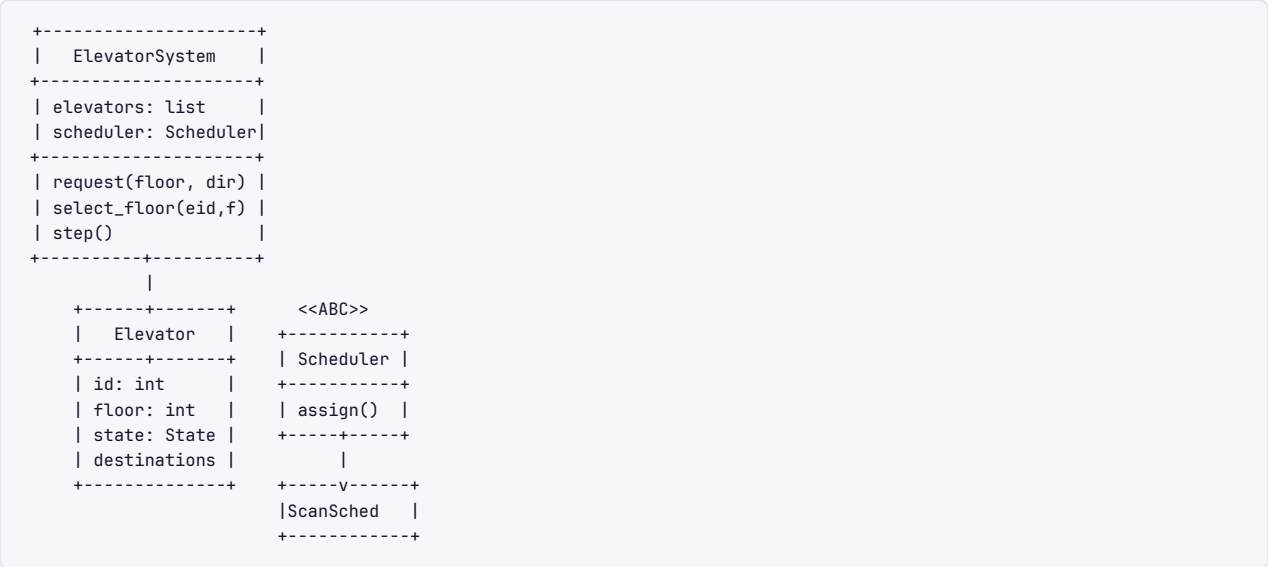
- How many elevators / floors? (Configurable.)
- Any elevator capacity / weight limit? (Mention, skip implementation.)
- Priority floors (emergency)? (Out of scope for now.)
- Which scheduling algorithm? (SCAN / ”look” algorithm — service in current direction first.)

- Real-time simulation or just state transitions? (State transitions.)

6.2 Core entities / classes

Class	Responsibility
`ElevatorState` (enum)	IDLE, MOVING_UP, MOVING_DOWN, MAINTENANCE
`Direction` (enum)	UP, DOWN, IDLE
`Request`	Floor + direction (external) or just floor (internal)
`Elevator`	Single car; state machine; queue of floors to serve
`Scheduler` (ABC)	Assigns external request to an elevator
`ScanScheduler`	SCAN/LOOK algorithm implementation
`ElevatorSystem`	Owns elevators + scheduler; top-level API

6.3 Class diagram (ASCII UML)



6.4 Design patterns used

- **State** — `ElevatorState` drives behavior. The transition table (`IDLE → MOVING_UP`, etc.) is explicit.
- **Strategy** — `Scheduler` is pluggable (SCAN, Nearest-Car, Zone-based).
- **Observer** (mention) — In production, elevators would publish state-change events; floors would subscribe to update displays.

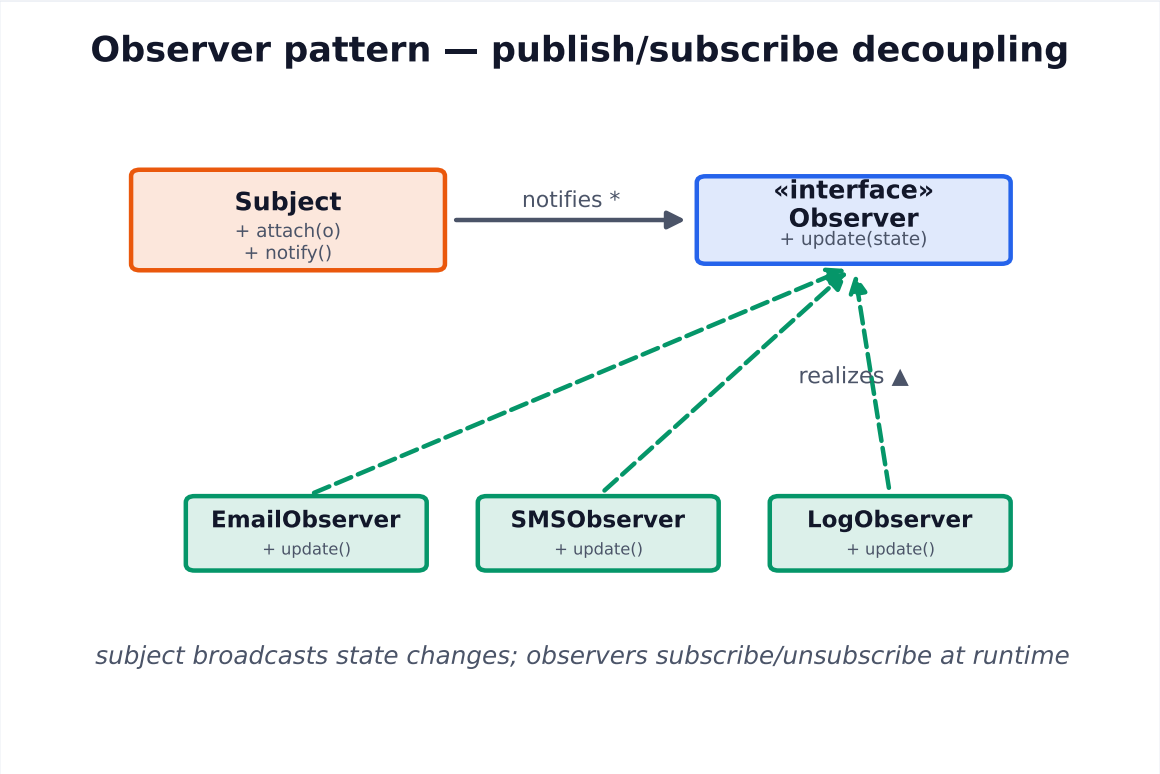


Figure 1. Observer lets a subject notify many observers of state changes without knowing their concrete types — the basis of event systems and reactive UIs.

6.5 Full code

Python

```python
from __future__ import annotations

from abc import ABC, abstractmethod
from enum import Enum, auto
from dataclasses import dataclass, field
from typing import Optional
import heapq


# ---------------------------------------------------------------- #
# Enums                                                            #
# ---------------------------------------------------------------- #

class ElevatorState(Enum):
    IDLE = auto()
    MOVING_UP = auto()
    MOVING_DOWN = auto()
    MAINTENANCE = auto()


class Direction(Enum):
    UP = auto()
    DOWN = auto()
    IDLE = auto()


# ---------------------------------------------------------------- #
# Request                                                          #
# ---------------------------------------------------------------- #
```

```python

@dataclass
class Request:
    floor: int
    direction: Direction  # Direction.IDLE for internal (in-elevator) requests
    is_internal: bool = False

    def __repr__(self) -> str:
        kind = "INT" if self.is_internal else f"EXT-{self.direction.name}"
        return f"Req(floor={self.floor},{kind})"


# ---------------------------------------------------------------------- #
# Elevator                                                               #
# ---------------------------------------------------------------------- #

class Elevator:
    """
    Single elevator car.

    Internally maintains two min-heaps:
      - up_queue: floors > current floor, to serve while going up (min-heap)
      - down_queue: floors < current floor, to serve while going down (max-heap via negation)
    SCAN algorithm: exhaust current direction first, then reverse.
    """

    def __init__(self, elevator_id: int, initial_floor: int = 1) -> None:
        self.elevator_id = elevator_id
        self.current_floor = initial_floor
        self.state = ElevatorState.IDLE
        self.direction = Direction.IDLE
        self._up_queue: list[int] = []    # min-heap
        self._down_queue: list[int] = []  # max-heap (store negatives)

    def add_destination(self, floor: int) -> None:
        if self.state == ElevatorState.MAINTENANCE:
            raise RuntimeError(f"Elevator {self.elevator_id} is under maintenance")
        if floor == self.current_floor:
            return
        if floor > self.current_floor:
            if floor not in self._up_queue:
                heapq.heappush(self._up_queue, floor)
        else:
            if -floor not in self._down_queue:
                heapq.heappush(self._down_queue, -floor)
        self._update_state()

    def step(self) -> Optional[int]:
        """
        Move one floor toward the next destination.
        Returns the floor reached, or None if already at destination / idle.
        """
        if self.state == ElevatorState.IDLE or self.state == ElevatorState.MAINTENANCE:
            return None

        if self.state == ElevatorState.MOVING_UP:
            if self._up_queue:
                next_floor = self._up_queue[0]
                self.current_floor += 1
                print(f"  Elevator {self.elevator_id} ↑ Floor {self.current_floor}")
                if self.current_floor == next_floor:
```

```
                    heapq.heappop(self._up_queue)
                    print(f"  Elevator {self.elevator_id} STOP at Floor {self.current_floor}")
            self._update_state()

        elif self.state == ElevatorState.MOVING_DOWN:
            if self._down_queue:
                next_floor = -self._down_queue[0]
                self.current_floor -= 1
                print(f"  Elevator {self.elevator_id} ↓ Floor {self.current_floor}")
                if self.current_floor == next_floor:
                    heapq.heappop(self._down_queue)
                    print(f"  Elevator {self.elevator_id} STOP at Floor {self.current_floor}")
            self._update_state()

        return self.current_floor

    def _update_state(self) -> None:
        """Transition state based on queues and current direction."""
        if not self._up_queue and not self._down_queue:
            self.state = ElevatorState.IDLE
            self.direction = Direction.IDLE
        elif self.direction == Direction.UP:
            if self._up_queue:
                self.state = ElevatorState.MOVING_UP
            else:
                # Exhausted upward requests; switch down
                self.direction = Direction.DOWN
                self.state = ElevatorState.MOVING_DOWN
        elif self.direction == Direction.DOWN:
            if self._down_queue:
                self.state = ElevatorState.MOVING_DOWN
            else:
                self.direction = Direction.UP
                self.state = ElevatorState.MOVING_UP
        else:
            # Was IDLE; pick direction based on first pending request
            if self._up_queue and (
                not self._down_queue or self._up_queue[0] >= self.current_floor
            ):
                self.direction = Direction.UP
                self.state = ElevatorState.MOVING_UP
            else:
                self.direction = Direction.DOWN
                self.state = ElevatorState.MOVING_DOWN

    def cost_to_serve(self, floor: int, direction: Direction) -> int:
        """
        Heuristic cost for scheduler: number of steps to reach `floor`.
        Considers current direction and queued stops.
        """
        if self.state == ElevatorState.MAINTENANCE:
            return float("inf")  # type: ignore[return-value]

        if self.state == ElevatorState.IDLE:
            return abs(self.current_floor - floor)

        if self.state == ElevatorState.MOVING_UP:
            if floor >= self.current_floor and direction == Direction.UP:
                return floor - self.current_floor
            # Must finish upward, come back
            top = max(self._up_queue) if self._up_queue else self.current_floor
```

```
            return (top - self.current_floor) + (top - floor)

        if self.state == ElevatorState.MOVING_DOWN:
            if floor <= self.current_floor and direction == Direction.DOWN:
                return self.current_floor - floor
            bottom = min(-x for x in self._down_queue) if self._down_queue else self.current_floor
            return (self.current_floor - bottom) + (floor - bottom)

        return abs(self.current_floor - floor)

    def __repr__(self) -> str:
        return (
            f"Elevator[{self.elevator_id}|Floor {self.current_floor}|"
            f"{self.state.name}|up={list(self._up_queue)}|"
            f"down={[-x for x in self._down_queue]}]"
        )


# ------------------------------------------------------------------ #
# Scheduler                                                          #
# ------------------------------------------------------------------ #

class Scheduler(ABC):
    @abstractmethod
    def assign(self, elevators: list[Elevator], request: Request) -> Elevator:
        ...


class NearestCarScheduler(Scheduler):
    """Assign the elevator with the lowest cost_to_serve heuristic."""

    def assign(self, elevators: list[Elevator], request: Request) -> Elevator:
        available = [e for e in elevators if e.state != ElevatorState.MAINTENANCE]
        if not available:
            raise RuntimeError("No elevators available")
        return min(
            available,
            key=lambda e: e.cost_to_serve(request.floor, request.direction)
        )


# ------------------------------------------------------------------ #
# Elevator System                                                    #
# ------------------------------------------------------------------ #

class ElevatorSystem:
    def __init__(
        self,
        num_elevators: int,
        num_floors: int,
        scheduler: Optional[Scheduler] = None,
    ) -> None:
        self.num_floors = num_floors
        self.elevators = [Elevator(i + 1) for i in range(num_elevators)]
        self.scheduler = scheduler or NearestCarScheduler()

    def external_request(self, floor: int, direction: Direction) -> None:
        """Someone on `floor` pressed the UP or DOWN button."""
        self._validate_floor(floor)
        request = Request(floor=floor, direction=direction)
        elevator = self.scheduler.assign(self.elevators, request)
```

```
        elevator.add_destination(floor)
        print(f"  Assigned {request} to Elevator {elevator.elevator_id}")

    def internal_request(self, elevator_id: int, floor: int) -> None:
        """Passenger inside elevator presses button for `floor`."""
        self._validate_floor(floor)
        elevator = self._get_elevator(elevator_id)
        elevator.add_destination(floor)
        print(f"  Elevator {elevator_id}: internal request for floor {floor}")

    def step_all(self, steps: int = 1) -> None:
        """Simulate `steps` time ticks."""
        for _ in range(steps):
            for elevator in self.elevators:
                elevator.step()

    def status(self) -> None:
        print("\n--- Elevator Status ---")
        for e in self.elevators:
            print(f"  {e}")

    def _validate_floor(self, floor: int) -> None:
        if not (1 <= floor <= self.num_floors):
            raise ValueError(f"Floor {floor} out of range [1, {self.num_floors}]")

    def _get_elevator(self, elevator_id: int) -> Elevator:
        for e in self.elevators:
            if e.elevator_id == elevator_id:
                return e
        raise ValueError(f"Elevator {elevator_id} not found")


if __name__ == "__main__":
    system = ElevatorSystem(num_elevators=2, num_floors=10)
    system.status()

    print("\n--- Requests ---")
    system.external_request(5, Direction.UP)   # person on floor 5 wants to go up
    system.external_request(3, Direction.DOWN)  # person on floor 3 wants to go down
    system.internal_request(1, 8)              # inside elevator 1, press 8

    system.status()

    print("\n--- Simulating 10 steps ---")
    system.step_all(steps=10)
    system.status()
```

6.6 Extensibility discussion

- **New scheduler (zone-based):** Implement `ZoneScheduler(Scheduler)` that restricts each elevator to a floor range. Inject at construction.
- **Emergency mode:** Add `ElevatorState.EMERGENCY`. Override `_update_state` to route all cars to ground floor first.
- **Capacity tracking:** Add `current_load: int` and `max_capacity: int` to `Elevator`. Scheduler skips full elevators.
- **Floor display / observer:** Elevators publish `FloorReachedEvent`; `FloorDisplay` objects subscribe and update arrow indicators.

Interview takeaway: The SCAN algorithm (also called ”elevator algorithm”) is exactly

how real elevator schedulers work — it's the same algorithm used by disk schedulers. Name it. The state machine with direction + two queues (up/down) is the key implementation insight.

7 6. Tic-Tac-Toe (Clean OO Board Game)

7.1 Requirements

Functional

- Two players, configurable board size (default 3×3).
- `make_move(player, row, col)` — place mark; validate; check win/draw.
- Win condition: any row, column, or diagonal fully owned by one player.
- Draw: board full, no winner.
- Support pluggable players (human, AI/random).

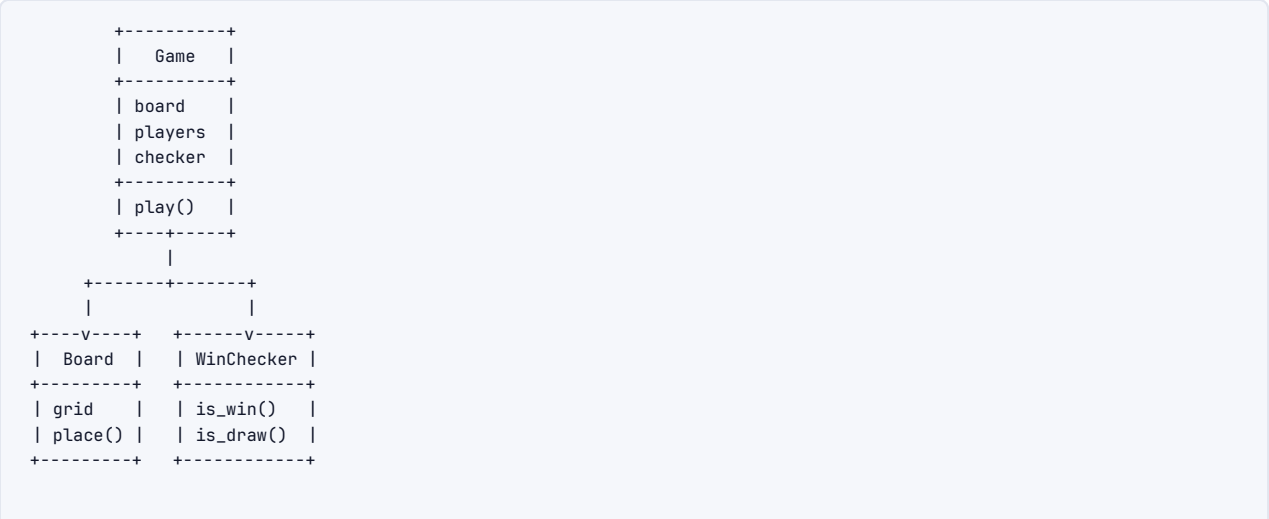
Clarifying questions to ask

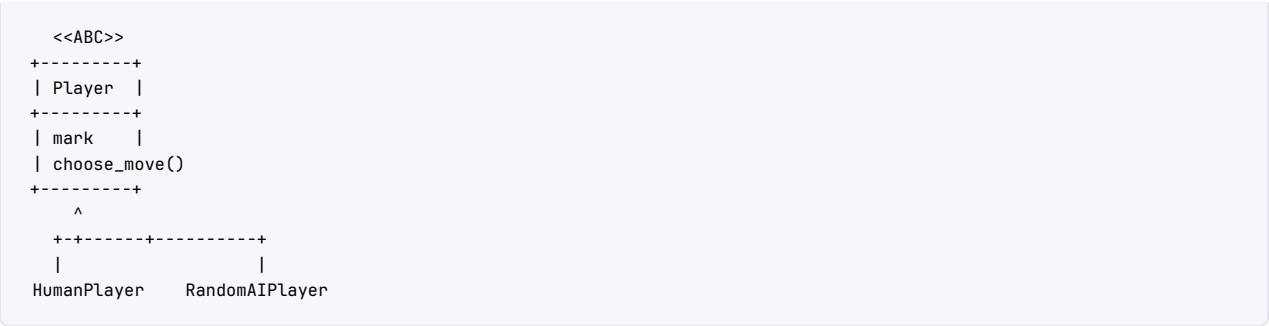
- Board size always 3×3 or configurable? (Configurable; win requires N-in-a-row.)
- Two players only, or more? (Two for now; mention extension.)
- AI player? (Optional; use Strategy.)
- Can the same player go twice? (No — enforce turn order.)
- Replay / undo moves? (Out of scope; mention Command pattern for undo.)

7.2 Core entities / classes

Class	Responsibility
`Mark` (enum)	X, O, EMPTY
`Board`	Grid state; place/validate mark; check win/draw
`Player` (ABC)	Name, mark, `choose_move(board) → (row, col)`
`HumanPlayer`	Reads from provided input (injectable for testing)
`RandomAIPlayer`	Picks a random empty cell
`WinChecker`	Encapsulates win-detection logic (single responsibility)
`Game`	Orchestrates turn loop; holds players + board

7.3 Class diagram (ASCII UML)





7.4 Design patterns used

- **Strategy** — `Player.choose_move` makes each player type a strategy for move selection.
- **Template Method** — `Game.play()` defines the fixed turn loop; player-specific behavior is deferred to `choose_move`.
- **Single Responsibility** — `WinChecker` is separate from `Board` (board manages state; checker evaluates it).
- **Command** (mention) — For undo, wrap each move in a `Command` object with `execute`/`undo`.

7.5 Full code

Python

```python
from __future__ import annotations

import random
from abc import ABC, abstractmethod
from enum import Enum, auto
from typing import Optional


# ---------------------------------------------------------------------- #
# Mark                                                                   #
# ---------------------------------------------------------------------- #

class Mark(Enum):
    EMPTY = "."
    X = "X"
    O = "O"


# ---------------------------------------------------------------------- #
# Board                                                                  #
# ---------------------------------------------------------------------- #

class Board:
    def __init__(self, size: int = 3) -> None:
        self.size = size
        self.grid: list[list[Mark]] = [
            [Mark.EMPTY] * size for _ in range(size)
        ]
        self._moves_made = 0

    def place(self, row: int, col: int, mark: Mark) -> None:
        if not (0 <= row < self.size and 0 <= col < self.size):
            raise ValueError(f"Position ({row},{col}) out of bounds")
        if self.grid[row][col] != Mark.EMPTY:
            raise ValueError(f"Position ({row},{col}) already occupied")
        self.grid[row][col] = mark
```

```
        self._moves_made += 1

    def is_full(self) -> bool:
        return self._moves_made == self.size * self.size

    def empty_cells(self) -> list[tuple[int, int]]:
        return [
            (r, c)
            for r in range(self.size)
            for c in range(self.size)
            if self.grid[r][c] == Mark.EMPTY
        ]

    def display(self) -> None:
        header = "  " + " ".join(str(c) for c in range(self.size))
        print(header)
        for r, row in enumerate(self.grid):
            print(f"{r} " + " ".join(cell.value for cell in row))
        print()


# ---------------------------------------------------------------- #
# Win Checker                                                      #
# ---------------------------------------------------------------- #

class WinChecker:
    """Stateless win-detection. Separated from Board (SRP)."""

    def is_winner(self, board: Board, mark: Mark) -> bool:
        size = board.size
        g = board.grid

        # Rows
        for row in g:
            if all(cell == mark for cell in row):
                return True
        # Columns
        for col in range(size):
            if all(g[row][col] == mark for row in range(size)):
                return True
        # Main diagonal
        if all(g[i][i] == mark for i in range(size)):
            return True
        # Anti-diagonal
        if all(g[i][size - 1 - i] == mark for i in range(size)):
            return True
        return False

    def is_draw(self, board: Board) -> bool:
        return board.is_full()


# ---------------------------------------------------------------- #
# Players                                                          #
# ---------------------------------------------------------------- #

class Player(ABC):
    def __init__(self, name: str, mark: Mark) -> None:
        self.name = name
        self.mark = mark

```

```python
    @abstractmethod
    def choose_move(self, board: Board) -> tuple[int, int]:
        ...

    def __repr__(self) -> str:
        return f"Player({self.name},{self.mark.value})"


class HumanPlayer(Player):
    """In tests, inject a moves iterator so no stdin needed."""

    def __init__(
        self,
        name: str,
        mark: Mark,
        moves_iter: Optional[iter] = None,  # type: ignore[type-arg]
    ) -> None:
        super().__init__(name, mark)
        self._moves_iter = moves_iter

    def choose_move(self, board: Board) -> tuple[int, int]:
        if self._moves_iter is not None:
            return next(self._moves_iter)
        while True:
            try:
                raw = input(f"{self.name} ({self.mark.value}), enter row col: ")
                row, col = map(int, raw.strip().split())
                return row, col
            except (ValueError, EOFError):
                print("  Invalid input. Enter two integers: row col")


class RandomAIPlayer(Player):
    def __init__(self, name: str, mark: Mark, seed: Optional[int] = None) -> None:
        super().__init__(name, mark)
        self._rng = random.Random(seed)

    def choose_move(self, board: Board) -> tuple[int, int]:
        empties = board.empty_cells()
        if not empties:
            raise RuntimeError("No moves available")
        choice = self._rng.choice(empties)
        print(f"  {self.name} plays ({choice[0]},{choice[1]})")
        return choice


# ---------------------------------------------------------------------- #
# Game                                                                   #
# ---------------------------------------------------------------------- #

class GameResult(Enum):
    WIN = auto()
    DRAW = auto()
    IN_PROGRESS = auto()


class Game:
    """
    Orchestrates the turn loop.
    Decoupled from I/O via injectable players.
    """
```

```python
    def __init__(
        self,
        player1: Player,
        player2: Player,
        board_size: int = 3,
    ) -> None:
        if player1.mark == player2.mark:
            raise ValueError("Players must have different marks")
        self.players = [player1, player2]
        self.board = Board(size=board_size)
        self.checker = WinChecker()
        self._current_idx = 0
        self.winner: Optional[Player] = None

    @property
    def current_player(self) -> Player:
        return self.players[self._current_idx]

    def play_turn(self) -> GameResult:
        player = self.current_player
        row, col = player.choose_move(self.board)
        self.board.place(row, col, player.mark)
        self.board.display()

        if self.checker.is_winner(self.board, player.mark):
            self.winner = player
            print(f"  {player.name} wins!")
            return GameResult.WIN

        if self.checker.is_draw(self.board):
            print("  It's a draw!")
            return GameResult.DRAW

        self._current_idx = 1 - self._current_idx
        return GameResult.IN_PROGRESS

    def play(self) -> Optional[Player]:
        """Run the full game loop. Returns winner or None on draw."""
        self.board.display()
        while True:
            result = self.play_turn()
            if result != GameResult.IN_PROGRESS:
                break
        return self.winner


if __name__ == "__main__":
    # Scripted demo: two AI players with fixed seeds for reproducibility
    p1 = RandomAIPlayer("Alice", Mark.X, seed=42)
    p2 = RandomAIPlayer("Bob", Mark.O, seed=99)

    game = Game(p1, p2, board_size=3)
    winner = game.play()
    if winner:
        print(f"Winner: {winner.name}")
    else:
        print("Game ended in a draw")

    # Human vs AI example (uncomment for interactive play):
    # human = HumanPlayer("You", Mark.X)
```

```
220     # ai = RandomAIPlayer("Bot", Mark.O)
221     # Game(human, ai).play()
```

7.6 Extensibility discussion

- **N-in-a-row win (Connect-4 style):** Parameterize `WinChecker` with `win_length < size`. Only the checker changes.
- **More than 2 players:** Change `self.players` to a list; cycle with `(current_idx + 1) % len(players)`. `Board` already supports arbitrary marks.
- **AI with Minimax:** Replace `RandomAIPlayer.choose_move` with a minimax search tree. The `Player` interface is unchanged.
- **Undo last move:** Wrap `board.place` in a `MoveCommand(Command)`. `game.undo()` calls `command.undo()` which clears the cell and decrements `_moves_made`.
- **Networked game:** `RemotePlayer(Player)` overrides `choose_move` to send the board state over a socket and wait for a response.

Interview takeaway: The WinChecker separation is a key OOP point — "Board manages state; WinChecker evaluates it." This satisfies SRP and makes unit testing trivial. Always mention the injectable moves iterator for testing HumanPlayer without stdin.

8 Common Patterns Across LLD Problems

8.1 Entity-extraction checklist

When you get a new problem, run through this in 5 minutes:

1. **Nouns → candidate classes:** Underline every noun. Parking Lot → ParkingLot, Level, Spot, Vehicle, Ticket, Fee.
2. **Verbs → candidate methods:** Circle every verb. "find available spot" → `find_spot(vehicle)`.
3. **Prune attributes:** Nouns that are just data fields (name, price, timestamp) become fields, not classes.
4. **Find the variability:** What changes? Pricing? Split type? Scheduling algorithm? That variability → Strategy pattern.
5. **Find the lifecycle:** What has state? Elevator (Idle → Moving), Ticket (Open → Closed) → State pattern or enum.
6. **Find the singleton:** What is there exactly one of? ParkingLot, ElevatorSystem → Singleton.
7. **Find the family:** Vehicles, Players, Strategies → Factory + ABC.

8.2 Which pattern fits which need

Need	Pattern	Example in this file
Swap algorithm at runtime	**Strategy**	Pricing, Split, Scheduler, Rate Limiter, Player
Object with lifecycle phases	**State**	Elevator (IDLE/MOVING/MAINTENANCE)
One global instance	**Singleton**	ParkingLot, ElevatorSystem
Create family of objects	**Factory Method**	VehicleFactory
Fixed algorithm, variable steps	**Template Method**	Vehicle.allowed_spot_types, Game.play
Undo / replayable actions	**Command**	Board moves (TicTacToe extension)

Need	Pattern	Example in this file
Notify many on change	**Observer**	ElevatorSystem → FloorDisplay
Wrap object with extra behavior	**Decorator**	Rate limiter middleware layer

8.3 Revision cheat sheet

```
LRU Cache        → HashMap + DLinkedList + sentinel head/tail
Parking Lot      → Strategy(pricing) + Factory(vehicle) + Singleton(lot)
Rate Limiter     → Strategy(algorithm) + per-client state
Splitwise        → Strategy(split) + greedy debt simplification
Elevator         → State machine + Strategy(scheduler) + SCAN algorithm
Tic-Tac-Toe      → Strategy(player) + SRP (Board vs WinChecker)
```

8.4 The five questions interviewers ask at the end

1. ”How would you make this thread-safe?” → Locks per shared resource; mention `threading.Lock` / `concurrent.futures`.
2. ”How would you scale this to 1M users?” → Move to Redis/DB-backed storage; horizontal sharding; mention CAP theorem for rate limiter.
3. ”Add feature X — what changes?” → Should be ”add a new class, zero changes to existing” (OCP). If it isn't, your design has a seam problem.
4. ”How would you test this?” → Unit test each strategy independently; mock dependencies; inject test doubles (e.g., moves iterator in TicTacToe).
5. ”What did you deprioritize?” → Show you know the tradeoffs: ”I skipped persistence, thread-safety on the outer loop, and error telemetry. For production I'd add X.”

Final interview takeaway: In a 45-min round, a clean, running 200-line solution with 4 well-named classes beats a 500-line ”enterprise” solution with 12 interfaces that doesn't run. Write the smallest correct design, name the patterns you used, and show one extensibility path. That's the formula.